



AI-Based Personalized Crop Advisory and Risk Management System

Better Information • Smarter Decisions • Better Harvests



Crop Intelligence

Personalized crop recommendations a



Farmer Friendly

Multilingual and voice/text interaction



Smart Decisions

Weather, soil, irrigation and profit guid



Farmer Market

Direct farmer-to-farmer buying and se

Abstract

Overview of the proposed smart farming platform

- Kisan Dost is an AI-enabled agricultural assistance platform designed for small and marginal farmers.
- It combines crop advisory, weather, mandi prices, soil health, plant disease detection and farm economics in one application.
- The system uses a simple Streamlit interface with multilingual selection and voice/text interaction.
- Farmers can compare crop choices, estimate profit and risk, plan irrigation, maintain a crop diary and use a farmer marketplace.
- The modular design allows real-time APIs and trained machine-learning models to be added later.

Problem Statement

Major challenges faced by farmers



Weather Uncertainty

Changing rainfall and temperature affect crop decisions.



Market Uncertainty

Farmers may not have timely mandi price information.



Soil Decisions

Crop and input choices depend on soil and water conditions.



Crop Diseases

Early disease identification can be difficult without expert help.



Profit Risk

Yield, input costs and prices create uncertainty.



Digital Barriers

Complex interfaces and language barriers reduce adoption.

Objectives

Goals of Kisan Dost

- Provide personalized crop recommendations using farm, soil, water and climate factors.
- Provide weather, rainfall and crop-risk information for better planning.
- Provide mandi price insights and a farmer-to-farmer marketplace.
- Support plant disease identification from uploaded or camera-captured images.
- Estimate cultivation cost, expected yield, revenue, profit and break-even point.
- Recommend irrigation schedules, crop rotation and crop-protection actions.
- Improve accessibility through multilingual labels and voice/text interaction.

Proposed System

Integrated services available after farmer login

 **Home**

Farm summary and alerts

 **Weather**

Forecast and rainfall risk

 **Mandi**

Market price insights

 **Crop Advice**

Crop match and risk

 **Soil Health**

pH and NPK guidance

 **Plant Doctor**

Disease image assistance

 **Crop Protection**

Pest and protection guidance

 **Irrigation**

Water recommendations

 **Cost & Profit**

Economics and break-even

 **Crop Diary**

Expenses and activities

 **Crop Rotation**

Next crop suggestions

 **What-If**

Scenario simulation

 **Agri Store**

Agricultural products

 **Buy & Sell**

Farmer marketplace

 **Dealers**

Nearby dealer directory

 **Kisan Forum**

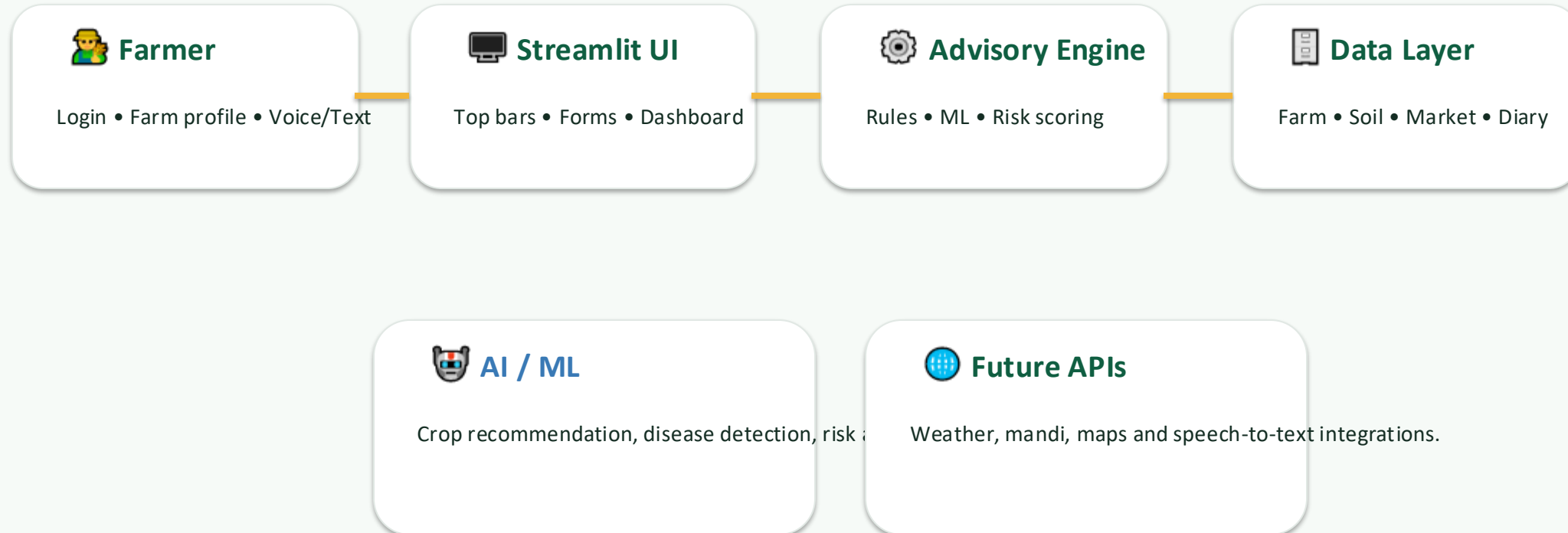
Farmer discussions

 **Schemes & News**

Government schemes and news

System Architecture

High-level application flow



Software Requirements

Development and runtime environment

 **Python**

Core programming language

 **Streamlit**

Web application and UI

 **Pandas**

Data processing

 **NumPy**

Numerical computation

 **Plotly**

Interactive charts

 **Pillow**

Image processing

 **ML / DL**

Scikit-learn, TensorFlow or PyTorch

 **IDE**

VS Code or PyCharm

Hardware Requirements

Recommended configuration



Processor

Intel i3 / Ryzen 3 or better



RAM

Minimum 4 GB; 8 GB recommended



Storage

5–10 GB free space



Camera

Useful for plant images



Microphone

For voice interaction



Internet

Recommended for real-time APIs

Implementation

Main development modules

1

Farmer Login

Mobile OTP / Registration ID and language selection.

2

Farm Profile

Location, acreage, soil, water source and crop details.

3

Advisory Engine

Processes farm information and generates recommendations.

4

Service Modules

Weather, mandi, soil, disease, irrigation and economics.

5

Marketplace

Buy/sell listings, prices and farmer directory.

6

Visualization

Cards, charts, alerts and what-if analysis.

AI / ML Components

Intelligent functions of the system

Crop Recommendation

Rank crops using soil, rainfall, temperature, water and farm conditions.

Disease Detection

CNN-based image classification can identify common crop diseases.

Risk Prediction

Combine weather, crop stage and market factors into a risk score.

Profit Prediction

Estimate revenue from yield and price ranges minus cultivation costs.

Expected Results

Benefits to farmers

- Faster access to important agricultural information in one place.
- More informed crop selection based on farm-specific conditions.
- Early awareness of plant disease symptoms through image-based assistance.
- Better irrigation and crop-protection planning.
- Improved understanding of cultivation cost, profit and break-even point.
- Better access to market information and direct farmer-to-farmer trading.
- A simple interface that can be adapted to regional languages and voice interaction.

Future Scope

Enhancements for a production-ready platform

Real-Time APIs

Live weather, rainfall and mandi data.

Trained AI Models

Validated recommendation and disease mod

Speech-to-Text

Automatic regional-language transcription.

Remote Sensing

Satellite/drone crop monitoring.

Precision Agriculture

GPS-based field mapping.

Secure Marketplace

Verified sellers, payments and order tracking

Smart Alerts

Personalized price, disease and irrigation ale

Mobile App

Android/PWA deployment for wider access.

Conclusion

Kisan Dost is an AI-based personalized crop advisory and risk management system developed to support farmers in making informed and timely agricultural decisions. The platform brings important services such as crop recommendation, weather and market-price information, soil health guidance, plant disease detection, irrigation planning, crop protection, cost and profit analysis, crop rotation, and a farmer-to-farmer marketplace into a single user-friendly application. By providing multilingual and voice/text-based interaction, the system aims to make digital agricultural assistance more accessible to small and marginal farmers. With future integration of real-time APIs, trained AI/ML models, remote sensing, GPS-based precision agriculture, and secure digital marketplace services, Kisan Dost can become a scalable smart farming solution that helps improve productivity, reduce risks, and support better farm profitability.

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THANK YOU



Kisan Dost

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